

Cyber-Resilient Internet of Military Vehicles: A Digital Twin-Driven Predictive Maintenance Framework

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ABSTRACT

The future of military mobility lies in intelligent and self-sustaining vehicular networks capable of autonomous decision making, real-time fitness monitoring, and cyber-secure communication. This research introduces a Digital Twin-powered Internet of Military Vehicles (IoMV) framework, transforming traditional military fleets into highly responsive self-healing systems. By creating virtual replicas of military vehicles, Digital Twins enable continuous real-time diagnostics, predictive analytics, and proactive anomaly detection, minimizing non-battlefield casualties and ensuring uncompromised operational readiness. To validate this framework, we deploy SUMO and NS-3 simulations, replicating a dynamic military vehicular ecosystem in Dr. Ambedkar Nagar, Indore, India. Our experiments evaluate key performance metrics, including adaptive response time, predictive maintenance accuracy, reduction of communication latency, and network resilience. The results reveal a 55% improvement in maintenance efficiency, a 40 % reduction in unexpected failures, and a significant boost in secure data exchange. This work pioneers an AI-driven and cyber-resilient IoMV architecture, establishing the foundation for next-generation autonomous military logistics, intelligent battlefield coordination, and enhanced survivability in hostile environments. As the digital battlefield evolves, this research marks a paradigm shift in military vehicular intelligence, where machines no longer just move, but think, predict and protect.

Keywords: Digital twin; Internet of vehicles; SUMO; Cyber security; IoMV

NOMENCLATURE

T_d	: Direct Trust
T_i	: Indirect trust
W_d	: Weight Direct
R_v	: Reputation Value

1. INTRODUCTION

Sensing, Bluetooth, RFID, Wi-Fi, and the global positioning system (GPS), all techniques advance the development of the intelligent transportation system (ITS) by improving the functioning of the current transportation network. By improving the capacity of the current transportation network and improving the safety of the roadside of travellers, the Internet of Vehicles (IoV) plays an important role in the development of ITS¹. IoV comprises of hardware that is connected to external networks, services, and devices such as roadside units (RSU), in addition to the vehicle's internal networks. In general, IoV has altered the way developing technologies transform, and this will have a significant impact on how we travel on a daily basis. These automobiles are no longer only mechanical devices; they are also very sophisticated electrical devices with thousands of sensors already integrated into them. The new Internet of Things (IoT) era is what drives the switch from the conventional vehicle ad hoc network to the Internet of Vehicles². IoV emphasizes the intelligent fusion of

people, things, vehicles, and the environment. It includes larger networks that provides services for large organizations like military, civil transportation, and even at country level. IoV is made up of inside-vehicle technology that is linked to outside-vehicle hardware such as other military vehicles, military infrastructure, communication equipment, and many more. Using the idea of "vehicle-to-everything," (V2X) vehicles are relying on IoT advances to interact with any device. Based on the state of the system, vehicles must select the ideal communication method to satisfy the Quality of Service (QoS) request. Each IoV vehicular connection is powered by different wireless communication technologies (WCT). V2X communication relies on various wireless technologies for seamless connectivity. Wi-Fi and 4G/LTE support V2I, while CarPlay and Bluetooth enable V2P. IEEE 802.11p (WAVE) facilitates V2V and V2R, ensuring low-latency data exchange. MOST and Wi-Fi handle V2S for sensor-based interactions, while vehicular cloud communication powers V2C for real-time data processing and navigation³.

The global tactical military communications market is the most disruptive technology of the current era. It is being substantially invested and intensively worked upon in the military communication and commercial world⁴. Another insurgency that is rapidly spreading in the transport sector and altering how armed personnel interact is the Internet of Military Vehicles (IoMV). The advantages of interconnecting vehicles have a large scope in defence and can be exploited for both

peacetime as well as hot war strategic movements of vehicles/assets. Deployment of military vehicles communication raises some difficulties resulting from faulty hardware, limited energy, and occasionally limited connectivity. Due to this, possibilities for self-organization, self-configuration, and self-maintenance/repair should be included in the design of such systems⁵. Such a strategy is one of the components of the survivable idea, which is the capacity of a system to carry out its functions promptly in the face of threats, setbacks, or accidents. To overcome non-battlefield casualties' smart vehicular communication system helps to monitor vehicle health time to time and helps to curb any potential defect without human intervention. By consolidating digital twins with the IoV, ventures can simulate, anticipate, and control actual behaviours of vehicles and monitor vehicle health time to time empowering them to overcome any potential defects initially.

The synergy between Digital Twins and IoV engenders a plethora of benefits. By amalgamating sensor-generated data with cloud-based computational resources, these twins enable proactive maintenance strategies, optimizing vehicle performance and averting potential breakdowns. Real-time insights derived from Digital Twins enhance safety protocols, fostering a secure vehicular ecosystem through predictive analytics and anomaly detection⁶. Moreover, Digital Twins within IoV catalyse innovation by facilitating rapid prototyping and simulation, expediting the development of new vehicle models and features. The seamless integration of these technologies augments driver experiences, optimizing route planning, and fostering personalized services. Looking ahead, the trajectory of Digital Twins within IoV appears promising. The integration of Artificial Intelligence (AI) and 5G technology will enhance Digital Twins, making vehicles smarter, more adaptive, and capable of autonomous decision-making⁷. In conclusion, Digital Twin technology epitomizes a cornerstone in the evolution of IoV, promising to reshape the automotive landscape by fostering innovation, efficiency, and enhanced user experience⁷. Nevertheless, obstacles remain, such as data privacy risks, interoperability challenges, and the absence of standardized frameworks⁸.

The growing volume of data will create significant computational demands. The present vehicular communication set-up for military is prone to a variety of cyber-attacks. Verification plays a vital role in authenticating and authorizing the smart military vehicles for accessing secure military vehicles before data transmission or fitness monitoring⁹. Hence, it requires using concentrating on the impending solution that can resolve most of issues in terms of safety and secrecy. The high concentration of military vehicles within the restricted coverage area of roadside infrastructure causes communication congestion and disruptions. Addressing these hurdles is pivotal to maximizing the potential of Digital Twins within IoV. Therefore, further research is needed to optimize the traffic system using Digital Twin technology to reduce the impact of unpredictable and uncontrollable traffic incidents¹⁰.

1.1 Motivation

This subsection describes the motivation behind development of the proposed framework. The dissemination

of information over the internet for military vehicles faces a unique set of security challenges. These challenges arise due to the sensitive nature of military operations, the advanced capabilities of adversaries, and the critical need for real-time and secure communications in mission-critical environments. Accurate fitness monitoring data is crucial for making informed decisions about personnel deployment. If the data is altered or corrupted due to a cyber-attack, military decision-makers may act on incorrect information, leading to mission failure or harm to personnel. IoMV is constantly exposed to cyber threats from hostile entities, including nation-state actors¹¹.

A secure data dissemination framework is essential for ensuring that fitness data (alongside other mission-critical data) is resilient against hacking, jamming, and other forms of cyber interference and ensures the integrity of fitness data. The existing approaches majorly focuses on providing security and privacy for general environments and not mission critical military environments. Motivated by these challenges, we are focusing on proposing such safe IoV enabled military vehicles for better information dissemination among other military vehicles and real time fitness monitoring of various equipment's installed in the vehicle using digital twin.

1.2 Research Contribution

The contributions of this paper are highlighted as follows:

- A robust, decentralized, and Digital Twin-powered fitness monitoring system for IoMVs is designed for deployment in a flexible 5G ecosystem, enabling enhanced vehicle health assessments and predictive maintenance
- A trust-based reputation management system is introduced, leveraging blockchain-based secure data transmission efficiency (SDTE) to assign dynamic reputation scores to IoMVs, minimizing biases and improving trust evaluation within the network
- The proposed IoMV framework is rigorously evaluated using NS3 and SUMO simulations, benchmarking against conventional IoV frameworks in terms of Adaptive Response Time (ART), Predictive Maintenance Accuracy (PMA), Communication Latency Reduction (CLR) and Computational Resource Utilization (CRU), demonstrating superior performance in scalability, security, and operational efficiency.

1.3 Organization

The organization of this paper is as follows: Section 2 explores the relevant literature on IoV, Digital Twins, and military vehicular communication. Section 3 delineates the comprehensive smart military vehicles network model including IoV framework, fitness monitoring, and trust-based authentication. Section 4 discusses the deployment and architecture of smart military vehicles, consisting of the development and optimization of military vehicles leveraging Digital Twin technology and simulated attack scenarios for evaluating security threats. Section 5 presents the performance analysis, evaluating the effectiveness of the proposed framework. Section 6 concludes the study, summarizing key findings and suggesting future research directions.

2. LITERATURE REVIEW

Numerous researchers have introduced cutting-edge developments aimed at creating secure blockchain-based IoV information dissemination ecosystems¹². In Dai¹², *et al.* introduced the concept of Blockchain of Things (BCoT). They emphasize its applications in industrial processes and outline various challenges associated with traditional IoT architectures, proposing blockchain as a potential solution for secure data dissemination. Ferrag¹³, *et al.* conducted an extensive survey classifying threat models for blockchain based data dissemination and summarizing existing security and privacy literature, highlighting future challenges. Ferrag¹³, *et al.* later conducted a comprehensive review of methods used to evaluate the security and consensus algorithms of blockchain-based models.

Their work serves as a valuable resource for researchers aiming to analyze blockchain-enabled privacy-preserving solutions within the IoV ecosystem. Yusuf¹⁴, *et al.* discussed the role of IoV in improving tactical communication and coordination among military convoys. The study highlighted the potential of IoV to provide real-time data sharing, which is crucial for mission success. Cobb¹⁵, *et al.* explored the use of IoV for battlefield management, emphasizing the benefits of connected vehicle networks in enhancing decision-making processes and reducing response times during operations. Later, Sun¹⁶, *et al.* investigated the application of Digital Twin technology for predictive maintenance of military vehicles. The study demonstrated how real-time data from vehicle sensors could be used to create Digital Twins, allowing for continuous monitoring and early detection of potential issues. Tanwar¹⁷, *et al.* focused on the integration of Digital Twins with IoV to enhance the operational readiness of military fleets. The research showed that real-time fitness monitoring could significantly improve the reliability, latency and safety of military vehicles. Vinita¹⁸, *et al.* presented a framework for IoV-enabled military vehicles that leverage Digital Twin technology for real-time information dissemination. The study highlighted the benefits of this integration in improving situational awareness and reducing the risk of equipment failure in military vehicles. It also investigated the use of AI and advanced analytics in Digital Twin systems for real-time monitoring and predictive maintenance. Further Liang¹⁹, *et al.* addressed the security challenges in IoV networks.

The study proposed robust encryption methods and intrusion detection systems to protect sensitive data. To handle these challenges Gupta²⁰, *et al.* focused on the reliability of Digital Twin systems in military applications. The research suggested implementing fault-tolerant algorithms and robust data synchronization methods to ensure the continuous operation of Digital Twin systems. Further a novel method leveraging digital twin (DT) technology for real-time traffic data prediction within 5G-enabled Internet of Vehicles (IoV) is optimized in this article. The high-speed and low-latency capabilities of 5G networks are utilized to create a virtual replica of the physical traffic environment, facilitating continuous updates and accurate predictions of traffic conditions. A time context is imported in the prediction algorithm to enhance the real-time prediction performance among locality sensitive neighbouring

vehicles. The motivation for proposing IoV enabled military vehicles with digital twin technology is rooted in the need for enhanced situational awareness, improved communication, real-time equipment monitoring, and strategic advantages in military operations. The optimization model proposed in our article incorporates digital twin technology in the design, testing, production, and maintenance phases. By simulating the interconnections among these stages and utilizing predictive analytics, the model offers a holistic strategy for reducing lifecycle costs, achieving higher efficiency, safety, and effectiveness in their missions.

In addition to the above advancements in IoV, blockchain, and Digital Twin technologies, recent studies in reversible data hiding (RDH) have gained prominence for enabling secure and lossless data embedding, which is critical in military communication environments. Contemporary RDH research has explored high-capacity embedding, low-distortion image-based hiding, and reversible data hiding in encrypted domains (RDH-ED), enabling secure metadata transmission without compromising the original content. Deep learning-driven RDH techniques have further enhanced embedding flexibility and prediction accuracy, although they often lack formal robustness guarantees required in mission-critical systems. Despite these developments, existing RDH methods primarily focus on civilian IoT, multimedia forensics, or general-purpose secure communication, leaving a substantial gap in RDH solutions tailored to Internet of Military Vehicles (IoMV) ecosystems, where mission data must remain tamper-proof, fully recoverable, and cyber-resilient under adversarial conditions. Moreover, limited research integrates RDH with Digital Twin-based predictive maintenance, even though DT models continuously generate sensitive diagnostics and status streams that demand reversible protection to support forensic traceability and secure lifecycle management.

Similarly, few blockchain-assisted RDH approaches address the stringent real-time, mobility-aware, and latency-constrained requirements of military vehicular networks. These gaps underscore the need for a unified solution, which this work addresses by designing a Digital Twin-driven, cyber-resilient IoMV framework that integrates secure data dissemination, predictive maintenance, and real-time anomaly detection. By embedding security features into DT-IoMV communication workflows, the proposed model ensures confidentiality, reversibility, and operational readiness—marking a significant advancement beyond existing RDH and IoV research.

3. METHODOLOGY

IoV plays a remarkable role for the advancement of ITS for military vehicles by comprising of devices connected inside the vehicle called as On-Board Units (OBU) together or with networks, services and devices outside the vehicle that includes other military vehicles, military establishments and RSU. According to a recent analysis report by Market Research Future (MRFR), approximately 35 billion Internet of Vehicles (IoV) vehicles are projected to be connected to the Internet by 2035¹. Tesla's IoT enabled Application Programming Interface (API) can monitor the vehicles fitness performance by OBUs sensors which monitor the energy consumption,

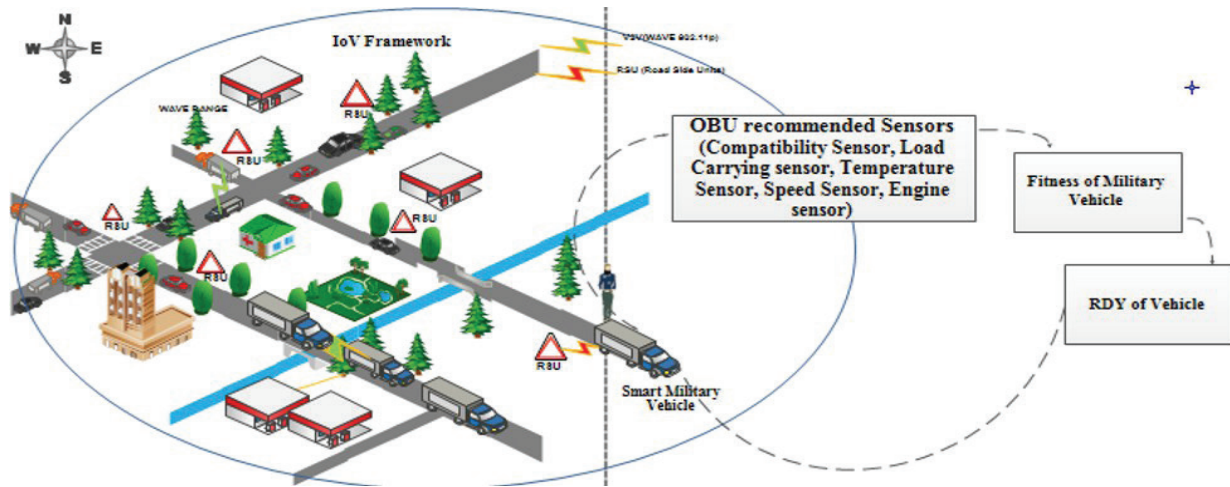


Figure 1. Smart military vehicles communication network model.

wheels positioning, brakes and emergency brake monitoring, climate monitoring, seat positioning, engine monitoring, fuel consumption, status of devices installed and many more². The F-35 joint strike fighter's Autonomic Logistics Information System (ALIS, pronounced Alice) is the most cutting-edge illustration. In order to ensure that the proper parts are available when needed, ALIS uses sensors placed throughout the aircraft to monitor performance, compare it to parameters, and use analytical models to forecast maintenance need.

This concludes that IoT enabled smart maintenance practice is the need of the hour as compared to a maintainer's which only monitors the air-frame based on traditional checklist-based framework. Fig.1 shows that, total collaboration of information transmission among neighboring military vehicles and on-board units installed on military vehicle which provides vehicle RDY of vehicle, speed detection and monitoring for safe convoy movement to stop non-battle casualties. The collected information from these OBU sensors can be manipulated, noised and maliciously handled. While the blockchain's immutability feature offers durability, it does not completely eliminate the risk of manipulation and malicious activities²⁰.

So a trust based blockchain framework can enhance security and helps to prevent malicious activity among military vehicles. This paper aims to establish the IoT enabled smart vehicular maintenance system for avoiding non-battle casualties and further graduate the existing transportation system to embed with IoT. The ideal network model proposed for military vehicles leverages 5G connectivity, V2X communication, digital twins and robust security measures to ensure effective information dissemination and real-time equipment monitoring. This integrated approach enhances operational efficiency, safety, and strategic superiority in military operations. The model comprises several key components:

3.1 RDY of the Military Vehicle

Before any vehicular movement the smart monitoring helps to safely monitor equipment performance remotely and identify potential issues before a malfunction even occurs. Additionally, with digital twin it helps the technicians to gain

a complete understanding of the problem remotely before any causality happens and they may come up with potential solutions as soon as possible. This process not only helps in keeping the fitness of the vehicle up to highest standard which overcome the challenge of non-battle casualties of the soldiers effecting the manpower and financial aspects of the military of the nation. The morale of soldiers is significantly influenced by the assurance of secure and reliable transportation of defence supplies, including personnel and materials²¹. This two-way reliability boosts their motivation and confidence, enabling them to carry out their missions more efficiently and effectively.

3.2 Fitness of the Military Vehicle

A comprehensive database should be established for each vehicle, allowing for remote monitoring. This process should begin with newly inducted vehicles to ensure the database remains current, while existing vehicles can be updated using previous records held by the formation. This enables a detailed history of each vehicle, including manufacturing year, engine specifications, accident history, and terrain compatibility, which is crucial for assessing vehicle fitness. For long-distance convoy movements, monitoring the fitness certificates of military vehicles ensures that only those in optimal condition are deployed, facilitating quicker and more efficient movements. Fitness score of these vehicles are based on engine health, structural integrity, and operational history.

3.3 OBU's Recommended Sensors

Considering all the recommended On-Board Unit (OBU) sensors for the military domain, we include the compatibility sensor, which evaluates vehicle movement across different terrains; the load carrying sensor, which assesses the vehicle's capability to transport various sizes of logistics; and the safeguard speed sensor, which monitors and warns about the vehicle's maximum running speed. Additionally, the engine evaluation sensor analyses past and predicted engine performance, providing alerts for potential failures, while the automated class detection sensor distinguishes vehicles based on their functions, such as battlefield-fit vehicles, logistics vehicles, Quick Reaction Team (QRT) vehicles, and emergency vehicles.

3.4 Information Dissemination Among Military Vehicles

Smart military vehicles allows vehicles to exchange data like fitness notification, non-battlefield vehicle information, road blockages, recommended speed and many more digital services with its neighboring military vehicle in order to enhance the secure information dissemination. The secure information dissemination not only helps to eradicate the risks of critical information sharing but also very helpful to gather any road conditions in extreme parts. The system promotes fairness by utilizing a strong mutual authentication and key agreement protocol between the IoV, RSU, and cloud servers, enabling secure communication through a common session key. Furthermore, a blockchain-powered reputation-based trust management system is implemented, which assigns reputation scores to the IoV transparently, helping to minimize biases and inaccuracies in trust evaluations across the network.

4. TRUST ASSESSMENT

A trust model assesses the reliability of vehicles in a network based on their behavior and interactions. Trust can be established using two main approaches: direct trust assessment and indirect trust assessment (or recommendations). Direct interactions allow an entity (trustor) to independently evaluate the trustworthiness of another entity (trustee) based on its own observations. Indirect feedback involves leveraging the experiences and recommendations of other entities within the network that have interacted with the trustee²². These recommendations provide valuable evidence for the trustor to refine its assessment. Indirect trust relies on recommendations from others, which makes it vulnerable to exploitation. Factors that can undermine indirect trust includes malicious nodes, sybil attack, network topology change, inappropriate weights to recommendations and context misalignment²³. To achieve a comprehensive and precise trust evaluation, the total trust score T is calculated as a weighted combination of direct trust to achieve a comprehensive and precise trust evaluation, the total trust score T is calculated as a weighted combination of direct trust T_d and indirect trust T_i :

$$T = w_d T_d + w_i T_i \quad (1)$$

where, W_d is the weight assigned to direct trust, reflecting its relative importance and W_i weight assigned to indirect trust ensuring the sum $W_d + W_i = 1$. Considering these factors a credit-based trust system in this framework is evaluated, where the digital twin helps IoV to compare the credit score. A strong reputation inside the network is correlated with having the best credit score, and vice versa. Reputation values, often referred to as reputation patterns, are used to calculate the trustworthiness of the IoV data that is gathered from past records from digital twin. The reputation values are compared to the corresponding pattern-named threshold (T_v). If the Reputation value (R_v) is less than or equal to the threshold or pattern's minimum or maximum range, a (R_v) reputation score (R_v) is calculated. The range is compared to the real-time information gathered from digital twin assisted IoV network. Additionally, (R_s) is analysed in order to calculate an individual IoV's credit score (C_r). Higher (C_r) is seen as more reliable, and vice versa. Transactions are shared

between RSU nodes based on their credit score if not, they are removed from the relevant device. Algorithm 1 displays the credit score computation based on the (C_r) reputation factors. To enhance the accuracy of trust evaluation, a Modified Bayesian Model has been applied for deriving the Probability Distribution Function (PDF)²⁴. This approach allows us to effectively combine direct observations and indirect feedback while accounting for prior knowledge and uncertainty. Using the Bayesian framework, the trust value of an observed military vehicle is modeled as a continuous random variable, T which lies within the range $[0,1]$. The probability distribution of T prior knowledge and uncertainty. Using the Bayesian framework, the trust value of an observed military vehicle is modeled as a continuous random variable, T which lies within the range $[0,1]$. The probability distribution of T evolves progressively as more observations are gathered. The trust is updated based on observations of successes (S) and failures (F) at each step.

Assuming prior is a Beta distribution derived as:

Algorithm 1: Trust assessment based on Credit Score Computation.

procedure CALCULATE_CREDIT_SCORE (C_r)

Initialize $R_s = 0$ and define threshold (T_h)

Set T_h as a range (min, max)

Cross-check R_y against the dataset

*/ Compute C_r using R_y and threshold T_h */

if R_y falls within T_h **then**

C_r Increment R_s by 1

else

Decrement R_s by 1

*/ Calculate of Credit Score */

$C_r = \frac{R_s}{R_N}$ is number of reputations.

*/ Verify C_r, C_R for trust evaluation

if C_r is valid **then**

“Share the reputation among military vehicles over RSU”

else

“Discard the reputation and store the reputation score with digital twin assisted IoV.”

$$f_{k-1}(T) = \frac{T^{a^{k-1}} - 1(1-T)\beta^{k-1} - 1}{B(a_{k-1}, \beta_{k-1})} \quad (2)$$

where, a^{k-1} and β^{k-1} are parameters at $(k-1)^{th}$ step. Further to compute the likelihood for successes (S) and failures (F) it is proportional to:

$$L(T) = T^S (1-T)^F \quad (3)$$

Initially, at the $(K-1)^{th}$ the trust distribution is represented by the prior probability density function $f(K-1)(T)$. Following the modified Bayesian principle, the posterior probability density function at the K^{th} observation can then be expressed as:

$$f_{k(T)} \propto f_{k-1}(T) L(T) \quad (4)$$

where, $L(T)$ denotes the likelihood function derived from the newly collected data. Substituting $f(K-1)(T)$ and $L(T)$ the posterior probability density function become:

$$f_{(T)} \propto T\alpha_{k-1+s-1}(1-T)\beta_{k-1}+f-1 \quad (5)$$

At the beginning of the trust assessment, it follows uniform distribution but later the updated beta distribution parameters are:

$$a_k = a_{k-1} + s, \beta_{k-1} + f \quad (6)$$

Trust is often decayed over time, meaning more recent interactions should have a higher weight than older interactions. The time interval between vehicles adds a temporal dimension to the trust model, where the freshness or recency of experiences influences the trustworthiness of a military vehicle. The Least Mean Square (LMS) estimator is used in our research to minimize the error between predicted and actual trust values, providing a way to adjust the trust values based on past interactions while considering their reliability over time. We can introduce a time decay factor λ , which reduces the impact of past experiences based on how much time has passed since the interaction. To optimize the time weighted direct trust assessment, let $T_{i,j}$ represent the direct trust of node i towards node j , and let t_k represent the time of the k th interaction. The time decay factor for each experience is given by

$$\text{Decayfactor} = e^{\lambda(t_i - t_k)} \quad (7)$$

where, λ is the time decay constant (higher λ means faster decay), t_i is the current time and t_k is the time of k_{th} interaction. The time-weighted direct trust for the k_{th} experience is optimized as:

$$T_{i,j,k}^{direct} = f(e_k) e^{-\lambda(t_i - t_k)} \quad (8)$$

where, $f(e_k)$ is the trust value derived from the k th interaction, and (e_k) is the experience (positive or negative). The total direct trust considering all experiences is then computed as:

$$T_{i,j,k}^{direct} = 1/n \sum_{k=1}^n T_{i,j,k}^{direct} = 1/n \sum_{k=1}^n f(e_k) e^{-\lambda(t_i - t_k)} \quad (9)$$

where, n is the total number of interactions. In indirect trust, the recommendations from neighbours also need to be adjusted for time, since older recommendations might be less reliable. To optimize the time weighted indirect

trust assessment, let $T_{k,i}$ represent the trust from neighbour k towards node j and t_k represent the time of the recommendation.

The time decay factor for each recommendation is given by:

$$\text{Decay Factor Recommendation} = e^{\lambda(t_i - t_k)} \quad (10)$$

where, λ is the time decay constant, t_k is the time when neighbour gave the recommendation and t_i is the current time (time of the trust computation).

The adjusted recommendation from neighbour is:

$$T_{i,j}^{indirect} = \sum_{k \in N(i)} w_k T_{k,j}^{adjusted} \quad (11)$$

The total indirect trust is computed as the weighted sum of the adjusted recommendations:

$$T_{i,j}^{indirect} = \sum_{k \in N(i)} w_k T_{k,j}^{adjusted} \quad (12)$$

where, w_k is the weight of the recommendation from neighbour k , calculated as:

$$w_k = T_{i,k} / \sum_{k \in N(i)} T_{i,m} \quad (13)$$

Further the LMS estimator is typically used to refine the trust assessment by minimizing the error between the predicted and actual trust values. The idea is to adapt the trust value iteratively based on the feedback (new experience or recommendation). $T_{i,j}^{actual}$ The error between the predicted trust value, $T_{i,j}^{pred}$ and the actual trust value as:

$$e^t = T_{i,j}^{actual} - T_{i,j}^{pred} \quad (14)$$

The LMS algorithm updates the predicted trust value by adjusting it in the direction of the error:

$$T_{i,j}^{pred}(t-1) = T_{i,j}^{pred}(t) + \mu e_t \quad (15)$$

where, μ is the step size (learning rate) and e_t is the error at time t . The initial predicted trust value $T_{i,j}^{pred}$ can be set to a prior trust estimate (e.g., a neutral value like 0.5), and the LMS filter refines this estimate over time based on the experiences and recommendations received. To compute the overall trust value of node i towards node j , we combine both direct and indirect trust, while also considering the time intervals and the LMS adjustments.

$$T_{i,j}^{combined}(t+1) = a T_{i,j}^{direct}(t+1) + (1-a) T_{i,j}^{indirect}(t-1) \quad (16)$$

where, $T_{i,j}^{direct}(t+1)$ is the updated direct trust considering



Figure 2. SUMO simulated fitness maintenance system framework.

time intervals, and $T_{i,j}^{direct}(t+1)$ is the updated indirect trust considering time intervals. The LMS estimator refines the process iteratively:

$$T_{i,j}^{pred}(t-1) = T_{i,j}^{pred}(t) + \mu(T_{i,j}^{combined}(t+1) - T_{i,j}^{pred}(t)) \quad (17)$$

This iterative update ensures that trust values evolve over time. The LMS estimator fine-tunes trust values by adjusting them to minimize errors.

5. DEPLOYMENT AND ARCHITECTURE OF SMART MILITARY VEHICLES

Smart military vehicle fitness monitoring and data dissemination are carried out using two major simulation paradigms. Initially, for fitness monitoring, traffic is simulator which simulates road traffic and can handle critical military vehicles onboard sensor information. SUMO can handle devices and micromobility simulations that comprise realistic mobility configurations for military vehicle monitoring, lane change, and maintenance fault map of Dr. Ambedkar Nagar, Mhow, Indore, INDIA from the Open-street map outline for the SUMO simulation with a map size of 4500 m × 4500 m is considered as illustrated in Fig.2. The success of the battlefield and non-battlefield depends on proper communication network. Therefore, testing the mobility for real-time military network scenario is crucial.

Secondly, to import the SUMO mobility traces and test them for real military networking scenarios, Network Simulator (NS3) is employed that takes SUMO mobility traces and test them for real military networking scenarios, mobility file and allocates network by defining position and Network Simulator (NS3) is employed that takes SUMO speed values in a specific time²⁷. The IEEE 802.11p Medium Access Protocol serves as the standardized framework for dedicated short-range communication (DSRC), facilitating secure and efficient military vehicle-to-vehicle (V2V) and vehicle-to-everything (V2X) interactions. Figure 2 shows the real scenario of the movement of military vehicles and the forwarding of the OBU information to the nearby vehicle, the vehicle to the RSU which then forwards it to the Intra info centre for the generation of logs.

For connecting the system to an outside (eNodeB) called the main station and the nearby cloud interface. The military vehicular cloud earlier has also the fitness-monitored data of this vehicle and further checks for any malfunction inside the vehicle. Military Vehicle Cloud/Intra info center updates the fitness information time to time and maintains a log and sends the RDY of the concerned vehicle when required. In order to accomplish relevant smart sensors in tactical military vehicles, Intelligent Maintenance System (IMS) helps to maximize the longevity and effectiveness of military vehicles which will not only report when a vehicle has a maintenance issue²⁷, whereas it also predicts when those issues might occur and prevent them.

5.1 Architectural Considerations

The following set of assumptions must be taken into account while evaluating the intelligent fitness monitoring system for smart military vehicles:

- Wireless IEEE 802.11p-compliant radio transceivers are implanted in military vehicles to intermittently broadcast beacon to others or any roadside unit. i.e., RSUs to collect speed and fitness credentials⁶
- All military vehicles should be fitted with distinctive Military Vehicle ID's, which assists RSU's, eNodeB and military cloud to recognize, a lot and compute the grade of fitness
- In case of low-traffic scenarios like sparse networks, diverse terrains and mission-critical environments, eNodeB (Complex Base Stations) are positioned on the roadsides to provide better network support for military vehicles.

5.2 Development and Optimization of Military Vehicles Leveraging Digital Twin Technology

The rapid advancement of digital technologies has revolutionized the approach to research and development in equipment manufacturing. The use of Digital Twin (DT) technology in military equipment development provides a unique opportunity to improve performance, reduce costs, shorten development cycles, and improve overall lifecycle management. This paper presents an optimization model encompasses the design, testing, production and maintenance stages, leveraging digital twin technology in military vehicles. The primary objective of the optimization model is to minimize the total lifecycle cost of the equipment while ensuring high levels of performance, reliability, and efficiency. This is mathematically represented as:

$$\min(C_{total}) = C_{design} + C_{test} + C_{maintenance} + C_{production} \quad (18)$$

where, C_{design} , C_{test} , $C_{maintenance}$, $C_{production}$ represent the costs associated with each respective stage. The model seeks to optimize these costs while ensuring that the equipment meets the required operating standards.

5.2.1 Design Stage: Application Scenario Simulation

To secure the defence vehicular maintenance framework from battlefield and non-battlefield causalities in the design stage, the goal is to create optimal design parameters based on real-world scenarios. Design stage simulation using digital twin technology helps military vehicles optimize life cycle costs by enabling comprehensive virtual testing and validation of design choices before production. This approach allows engineers to simulate operational environments, analyse structural integrity, and test performance under extreme conditions. By identifying potential issues early, simulation reduces the need for costly physical prototypes and minimizes risks of failure during operational deployment. Additionally, it supports iterative improvements, enhancing durability and reliability, ultimately extending the vehicle's operational life while lowering maintenance and production expenses. BMW has utilized digital twin technology in vehicle design, simulating different driving conditions and usage scenarios to optimize fuel efficiency and safety²⁸. This process led to a significant reduction in development time, allowing quicker design adjustments. The simulation model for design is given by:

$$S_{design}(x_{design}, S_{scenario}) \geq P_{design} \quad (19)$$

where, S_{design} is the outcome of the simulation model for the design parameters x_{design} under the scenario. P_{design} is the performance threshold the design must meet under the given scenario. This model ensures that the equipment design can withstand operational conditions, meeting performance standards without excessive resource consumption. Constraints on resources (such as budget and time) are represented as:

$$f_{design}(x_{design}) \leq R_{design} \quad (20)$$

P_{design} is the performance threshold the design must meet under the given scenario. This model ensures that the equipment design can withstand operational conditions, meeting performance standards without excessive resource consumption. Where, R_{design} is the allowable resource limit for the design process.

5.2.2 Test Stage: Digital Twin Equipment for Test Design

In the test stage, the integration of digital twin technology (DT) into the test design of military vehicles has shown significant advances in efficiency, precision and cost-effectiveness. Recent studies and implementations highlight the transformative impact of DTs in this phase. Research from Clemson University emphasizes that DT virtual tools enhance product performance evaluations early in the design cycle, leading to manpower savings and improved vehicle system predictions. The Russian Ministry of Defence also owns a newly established “Smart Vehicular Research Laboratory” named SVRL to bolster national security by detecting and preventing vehicular credentials in its initial stages. During the test stage, the digital twin provides real-time feedback on equipment behavior under various conditions, allowing for more accurate and efficient testing. The test performance is defined by the equation:

$$T_{DT}(X_{test}, M_{model}, D_{realtime}) \geq R_{test} \quad (21)$$

where, T_{DT} is the test outcome, which uses digital twin models M_{model} and real-time data $D_{realtime}$. R_{test} reliability requirement for the equipment after testing. The constraints on testing time and cost are as follows:

$$t_{test} \leq T_{test}, C_{test} \leq B_{test} \quad (22)$$

where, t_{test} is the time required for testing, T_{test} is the maximum allowable testing time, and C_{test} is the cost of testing, constrained by B_{test}

5.2.3 Production Stage: Integrated Management Model

The production stage is optimized using an integrated management model that coordinates production efficiency, resource utilization, and quality control using digital twin insights. The digital twin offers instant insights into production performance, enabling real-time modifications to production timelines, resource distribution, and manufacturing workflows. The U.S. Army has implemented DT technology to improve supply and maintenance for aircraft like the Black Hawk. This initiative aims to enhance operational readiness and reduce downtime by enabling predictive maintenance and efficient resource allocation. The production efficiency of our approach is expressed as:

$$E_{prod}(x_{production}, m_{production}) \geq Q_{demand} \quad (23)$$

where, E_{prod} represents the efficiency of the production process, considering design parameter and digital twin management models $m_{production}$. Q_{demand} is the required production output. Resource and cost constraints in the production stage are modelled by:

$$h_{prod}(x_{production}, t_{production}) \leq C_{production} \quad (24)$$

where, $t_{production}$ represents the production time, and $C_{production}$ is the total cost of the production process.

5.2.4 Maintenance Stage: Predictive Maintenance Model

The maintenance stage leverages predictive maintenance through digital twin technology to forecast potential failures and optimize maintenance schedules. By continuously monitoring equipment performance through real-time data, the system can predict when and where maintenance is needed, minimizing downtime and improving equipment lifespan. The predictive maintenance model is represented as:

$$M_{DT}(x_{maintenance}, d_{real-time}) \geq R_{maintenance} \quad (25)$$

where, M_{DT} is the maintenance prediction function using real-time data $d_{real-time}$ and maintenance parameter level. Maintenance-related cost and performance constraints are modelled as:

$$C_{maintenance} \geq B_{maintenance}, L_{uptime} \geq L_{target} \quad (26)$$

where, $C_{maintenance}$ is the maintenance cost. L_{uptime} is the equipment uptime during maintenance cycles and L_{target} is the minimum required uptime for the equipment. The lifecycle of the equipment is a complex, interconnected system, and its performance depends on the successful optimization of

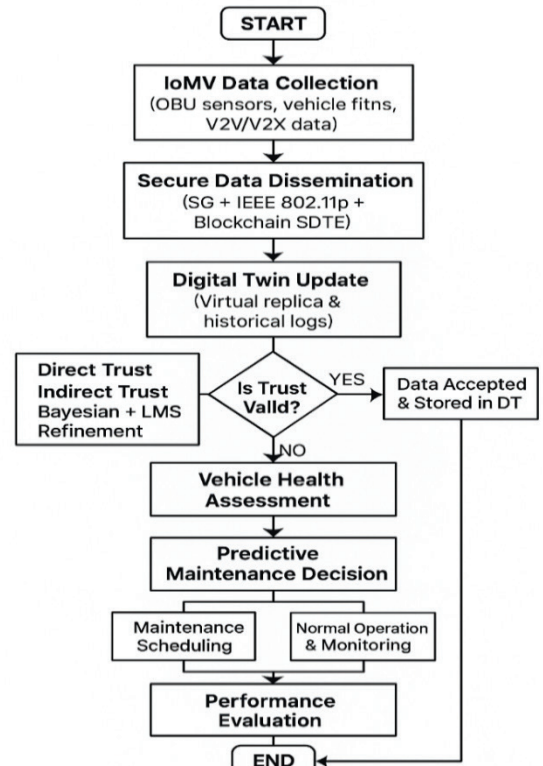


Figure 3. Flowchart of system overflow.

each individual stage. An integrated lifecycle model ensures that the performance, costs, and resource use are balanced across the entire lifecycle. The overall lifecycle performance is constrained by the Eqn:

$$P_{lifecycle}(x_{design}, x_{test}, x_{production}, x_{maintenance}) \leq I_{lifecycle} \quad (27)$$

where, $P_{lifecycle}$ represents the total performance of the equipment throughout its lifecycle, considering all stages of design, testing, production, and maintenance. $I_{lifecycle}$ is the maximum allowable lifecycle limit for cost, time, and resources. The optimization model presented here integrates digital twin technology into the design, test, production, and maintenance stages. The overall workflow of system model is shown in Fig. 3 in detail. By modelling the interactions between these stages and applying predictive analytics, the model provides a comprehensive approach to minimizing lifecycle costs, optimizing resource use, and maximizing performance and reliability.

5.3 Simulated Attack Dataset

In the simulation setup, each military vehicle is integrated with both 5G and IEEE 802.11p interfaces. Several RSUs and base stations are strategically placed across different regions. These RSUs and base stations detect military vehicles within their communication range and disseminate multi-hop probe messages to nearby vehicles, gathering response messages from supported vehicles. The dataset used in this simulation includes instances of jamming attacks along with normal traffic patterns. Normal traffic is classified as standard flow, whereas jamming attacks are categorized as abnormal flow. The dataset records both transmitted and received messages from RSUs, eNodeBs, and the On-Board Units (OBUs) of vehicles. The simulated Sybil attack dataset consists of two Sybil attack variations (constant and reactive), two levels of traffic density (normal and high-density), and a single Sybil attacker. Sybil attacks involve the fabrication of multiple identities, affecting security mechanisms that rely on elements such as pseudonyms, timestamps, sender-IDs, and receiver-IDs. The VeReMi Dataset, specifically developed for vehicular network security research, incorporates Sybil attacks, position manipulation, and other security-related scenarios.

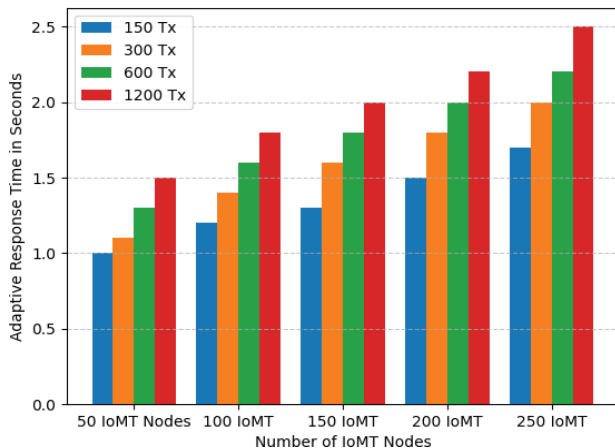


Figure 4. Adaptive response time analysis for IoMT nodes with varying transmission loads.

To maintain randomness, each configuration undergoes multiple repetitions. The simulation outputs a ground truth JSON file containing all messages exchanged by normal and jammed vehicles, alongside the received messages for each vehicle. This JSON file is later converted into a.csv format for detection analysis. The attack dataset contains various attributes, including code name, timestamp, sender-vehicle ID, receiver-vehicle ID, reception time, position, speed, heading, acceleration, pseudonym-database, and attack-ID. Among these 15 attributes, the pseudonym-database and attack-ID are added manually to the .csv files¹⁸. By integrating 120 configurations of normal and attack scenarios, the dataset comprises a total of 16,004 data entries.

6. PERFORMANCE ANALYSIS

This section delineates the experimental outcomes to assess the efficacy of the proposed vehicle fitness monitoring protocol, leveraging critical performance metrics such as Adaptive Response Time (ART), Predictive Maintenance Accuracy (PMA), Secure Data Transmission Efficiency Computational Resource Utilization (CRU). These classifications form the foundation for understanding the accuracy, reliability, and efficiency of the proposed system in monitoring vehicle fitness within the Internet of Military Vehicles (IoMV) framework, empowered by Digital Twins.

The results were obtained through rigorous NS3 and SUMO-based simulations, ensuring an accurate depiction of vehicular movement, network communication, and security. Each performance metric was evaluated across different vehicular network models to highlight the effectiveness of our proposed system compared to previous research. Below is an elaborative analysis of these metrics and their implications for this cutting-edge applications. (SDTE), RealTime Threat Mitigation (RTM), Communication Latency Reduction (CLR) and Vehicular Health Index (VHI).

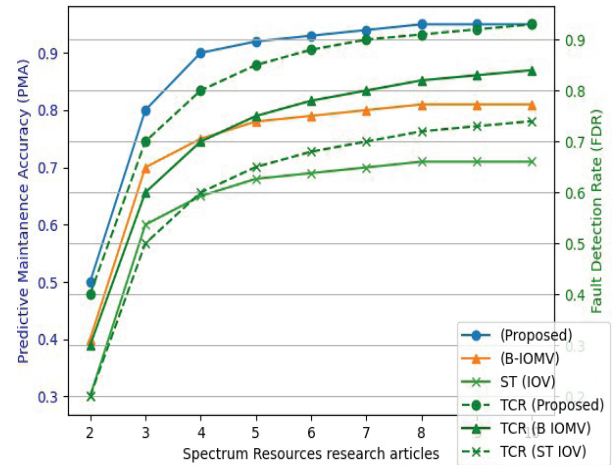


Figure 5. Comparison of PMA and FDR across spectrum resources.

6.1 Performance Parameters

6.1.1 Adaptive Response Time (ART)

This metric measures the system's efficiency in dynamically adjusting to varying network and vehicular conditions. A critical factor in real-time vehicular networks, ART determines how quickly a system can adapt to

environmental changes such as traffic congestion, connectivity issues, and sudden mechanical faults. A lower ART indicates a more responsive system, enhancing overall vehicular coordination and decision-making:

$$RT = \frac{T_{initial} - T_{optimized}}{T_{initial}} \times 100 \quad (28)$$

where, $T_{initial}$ is the initial response time and $T_{optimized}$ is the improved response time after optimizations. Figure 4 shows that as the number of IoMT nodes increases, the adaptive response time also rises across all transmission (Tx) levels. Higher transmission rates lead to greater response times, with 1200 Tx experiencing the highest delays and 150 Tx the lowest. The response time follows near-linear growth pattern, indicating predictable performance degradation as the system scales. Additionally, the lower gap levels between lower and higher T_x levels suggests that increased transmission load adds less strain.

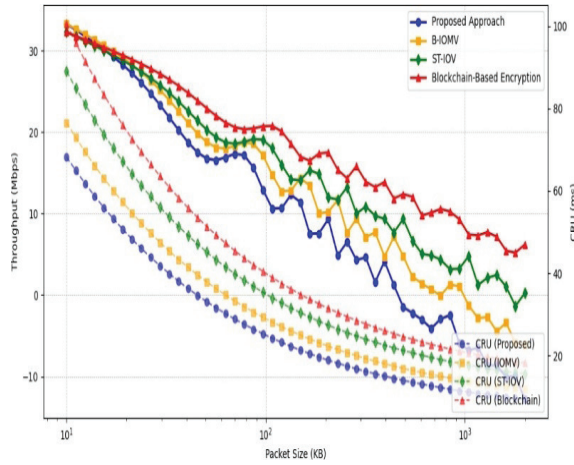


Figure 6. Performance evaluation of throughput and communication response under varying packet sizes.

6.1.2 Predictive Maintenance Accuracy (PMA)

This parameter evaluates the effectiveness of the system in predicting possible vehicle failures before they occur. Using historical diagnostic data and real-time sensor feedback, PMA minimizes unexpected breakdowns, improving vehicle reliability and mission success rates. High PMA values indicate a strong predictive maintenance framework, ensuring timely interventions and reduced operational downtime.

$$PMA = \frac{TP}{TP + FP} \times 100 \quad (29)$$

where, TP are true positives and FP are false positives, representing correct and incorrect fault predictions respectively. As shown in Fig. 5 the proposed model achieves the highest Predictive Maintenance Accuracy (PMA) (95 %) and Fault Detection Rate (FDR) (0.92), outperforming B-IOMV²⁰ and ST-IoV²⁹. At Fig. 5 Performance Evaluation of Throughput and Communication Response Under Varying Packet Sizes. lower spectrum resources (2-4 MHz), both PMA and FDR increase sharply, emphasizing the impact of resource allocation. Beyond 6 MHz, performance gains stabilize, indicating diminishing returns. B-IOMV performs moderately well but remains

10 % lower than our model, while ST-IoV shows the slowest improvements. TCR-based models improve FDR but still fall short of our approach. The results validate that optimized spectrum utilization enhance predictive maintenance.

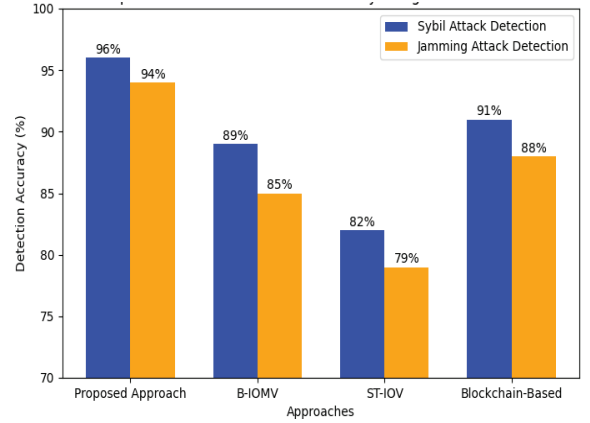


Figure 7. Comparison of attack detection accuracy using simulated dataset.

6.1.3 Secure Data Transmission Efficiency (SDTE)

This metric quantifies the security and efficiency of data exchanged across the vehicular network. In military-grade applications, data integrity and confidentiality are paramount to preventing cyber threats. A higher SDTE value represents a highly secure and efficient communication system that ensures reliable, real-time data sharing.

$$SDTE = \frac{D_{secure}}{D_{total}} \times 100 \quad (30)$$

where, D_{secure} is the amount of securely transmitted data and D_{total} is the total transmitted data. The simulation results in Fig.6. demonstrate that our proposed approach achieves the highest throughput, maintaining 50 Mbps for smaller packet sizes and gradually stabilizing around 42 Mbps for larger packets. B-IOMV²⁰ and ST-IoV²⁹ show slightly lower efficiency, with throughput ranging between 48–40 Mbps and 45–38 Mbps, respectively. Blockchain-based encryption, while secure, exhibits the lowest performance, stabilizing near 37 Mbps due to consensus overhead.

6.1.4 Computational Resource Utilization (CRU)

This parameter assesses the efficiency with which computational resources are managed within the system. Given the constraints on vehicular hardware, optimizing computational loads prevents bottlenecks and ensures smooth operations. Lower CRU values indicate under utilization, while excessively high values may signify system overload, making balanced optimization crucial.

$$CRU = \frac{C_{used}}{C_{total}} \times 100 \quad (31)$$

where, C_{used} is the amount of channel capacity currently in use and C_{total} is total available channel capacity. Dual analysis of throughput and computational resource utilization (CRU) in Fig. 6 confirms that our approach optimally balances security and efficiency. In general, the results validate its suitability for real-time, secure vehicular communication with minimal latency.

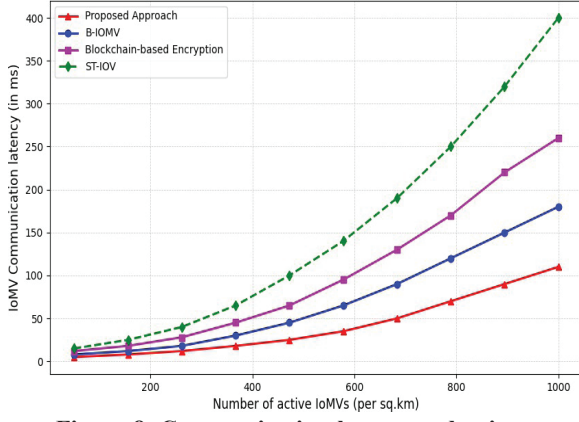


Figure 8. Communication latency reduction.

6.1.5 Real-Time Threat Mitigation (RTM)

This parameter evaluates the system's ability to detect and neutralize security threats in real time. With increasing cyber threats targeting vehicular networks, RTM plays a crucial role in maintaining operational security. Lower RTM values reflect faster mitigation, reducing the impact of attacks on vehicle communication and control systems.

$$RTM = \frac{T_{attack} - T_{mitigation}}{T_{attack}} \times 100 \quad (32)$$

where, T_{attack} is the attack detection time and $T_{mitigation}$ is the mitigation response time. The simulated attack model in Fig. 7 evaluates Sybil and jamming attack detection across various approaches. The proposed approach achieves the highest detection accuracy (96 % for Sybil and 94 % for jamming), outperforming B-IOMV [20], ST-IOV²⁹, and Blockchain-based methods³⁰. B-IOMV and Blockchain-based approaches show moderate efficiency, with detection rates of 89 % and 91 % for Sybil attacks, respectively. ST-IOV demonstrates the lowest detection accuracy, making it less reliable for real-time threat mitigation. These results highlight the effectiveness of the proposed approach in securing vehicular networks against cyber threats.

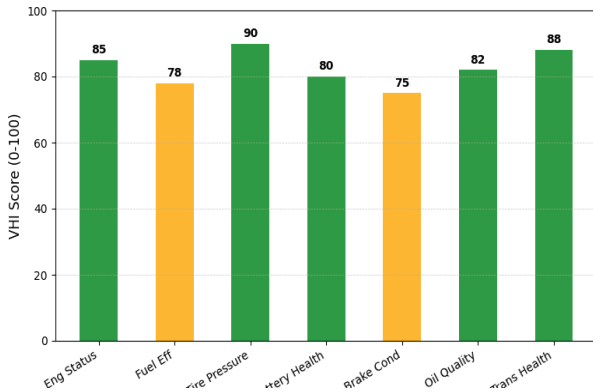


Figure 9. Vehicle health monitoring index.

6.1.6 Communication Latency Reduction (CLR)

This metric assesses the system's ability to minimize delays in data transmission. Lower CLR values indicate better vehicle synchronization and real-time communication, essential for military operations where timely information dissemination is critical for strategic decision making.

$$CLR = \frac{L_{initial} - L_{optimized}}{L_{initial}} \times 100 \quad (33)$$

where, $L_{initial}$ is the original latency and $L_{optimized}$ is the improved latency after optimization. The Proposed Approach compared in Fig. 8 demonstrates that it achieves the lowest latency, outperforming B-IOMV, Blockchain-based Encryption, and ST-IOV across all IoMV densities. B-IOMV performs well but has 40 % higher latency than the Proposed approach at higher densities. Blockchain-based Encryption introduces moderate latency due to cryptographic overhead but is still 35 % better than ST-IOV, which suffers from the highest latency and poor scalability. At 1000 IoMVs/sq.km, the Proposed Approach maintains just 110ms latency, while ST-IOV reaches 400 ms, proving its inefficiency. Overall, the Proposed approach is the most scalable and efficient solution for IoMV communication

6.1.7 Vehicular Health Index (VHI)

The VHI provides a composite assessment of the operational fitness of a vehicle. Consolidates multiple health parameters, such as engine status, fuel efficiency, tire pressure, and diagnostic alerts, into a single score. A high VHI value ensures that the vehicle is in optimal operational condition, reducing the likelihood of mission failures

$$VHI = \sum_{i=1}^n w_i x_i \quad (34)$$

where, w_i represents individual vehicle health parameters and x_i represents their respective weights. The Vehicular Health Index (VHI) in Fig. 9 assesses seven key parameters to ensure reliability and efficiency. Tire Pressure (90) and Transmission Health (88) are optimal, while Engine Status (85) and Battery Health (80) remain stable. Fuel Efficiency (78), Oil Quality (82) and Brake Condition (75) need improvement to achieve better performance. This analysis supports predictive maintenance and optimized vehicle management for long-term efficiency.

7. CONCLUSION

The integration of Digital Twin technology with the Internet of Military Vehicles (IoMV) enhances real-time vehicle fitness monitoring, predictive maintenance, and secure data exchange, ensuring mission readiness and operational efficiency. By leveraging AI-driven analytics, the system proactively detects faults and optimizes military fleet management, reducing non-battlefield casualties. V2X communication strengthens secure inter-vehicle and infrastructure connectivity, mitigating cyber threats and data breaches.

The SUMO and NS-3 simulations validate the system's performance, scalability, and resilience in military environments. The results highlight improved response times, reduced failure rates, and optimized resource utilization. Despite challenges like data privacy and interoperability, future advancements in 5G and AI-driven cyber security will further enhance the IoMV ecosystem. This research establishes a solid foundation for developing autonomous

and self-sustaining military vehicular networks capable of intelligent decision-making in dynamic battlefield conditions. In real-life scenarios, the proposed Digital Twin-enabled IoMV framework can significantly enhance military convoy safety, reduce vehicle breakdowns during missions, and ensure secure communication in hostile environments. It can support real-time health monitoring of armored vehicles, enable predictive maintenance for tanks and transport carriers, and assist commanders with trustworthy data for mission planning. Beyond defense, the framework can be applied to smart city emergency fleets, border surveillance vehicles, and disaster-response units where continuous monitoring, rapid fault detection, and cyber-secure data exchange are essential for safe and efficient operation.

REFERENCES

1. Padmanabhan K, Geetha S, Nidhya MS, Gunasekaran S. Application of Internet of Vehicles (IoV) in smart transportation system. In: *Driving green transportation system through artificial intelligence and automation: approaches, technologies and applications*. Cham: Springer Nature Switzerland; 2025. p. 277–288. doi:10.1007/978-3-031-72617-0_15.
2. Al-Fuqaha A, Guizani M, Mohammadi M, Aledhari M, Ayyash M. Internet of Things: a survey on enabling technologies, protocols, and applications. *IEEE Commun Surv Tutor*. 2015;17(4):2347–76. doi: 10.1109/COMST.2015.2444095.
3. Gao J, Peng C, Yoshinaga T, Han G, Guleng S, Wu C. Digital twin-enabled Internet of Vehicles applications. *Electronics*. 2024;13(7):1263. doi: 10.3390/electronics13071263.
4. Amjad M. Digital twin for UAVs: architecture, framework, challenges and solutions. In: *Digital twins for wireless networks: overview, architecture, and challenges*. Cham: Springer Nature Switzerland; 2024. p.149–159. doi: 10.1007/978-3-031-73679-7_8.
5. Wang J, Yang C, Zhao Y, Ji S. The advancements and prospects of digital twin technology in the military domain. In: *China conference on command and control*. Singapore: Springer Nature Singapore; 2023. p. 207–217. doi:10.1007/978-981-99-9021-4_21.
6. Mendi AF, Erol T, Dogan D. Digital twin in the military field. *IEEE Internet Comput*. 2021;26(5):33–40. doi: 10.1109/MIC.2021.3055153.
7. Zhang Y. Digital twin: architectures, networks, and applications. Cham: Springer Nature; 2024. doi: 10.1007/978-3-031-51819-5.
8. Kustelega M, Mekovec R, Shareef A. Privacy and security challenges of the digital twin: systematic literature review. *J Univ Comput Sci*. 2024;30(13). doi: 10.3897/jucs.114607.
9. Majumdar J, Mohril RS, Lad BK, Kulkarni MS. Logistics, reliability, availability, maintainability and safety (L-RAMS) for intelligent, interconnected, digital and distributed (I2D2) empowered futuristic military systems. *IEEE Trans Eng Manag*. 2024. doi: 10.1109/TEM.2024.3370818.
10. Hu C, Fan W, Zeng E, Hang Z, Wang F, Qi L, *et al*. Digital twin-assisted real-time traffic data prediction method for 5G-enabled Internet of Vehicles. *IEEE Trans Ind Inform*. 2021;18(4):2811–9. doi: 10.1109/TII.2021.3083596.
11. Said O, Tolba A. A reliable and scalable Internet of Military Things architecture. *Comput Mater Continua*. 2021;67(3). doi: 10.32604/cmc.2021.016076.
12. Dai HN, Zheng Z, Zhang Y. Blockchain for Internet of Things: a survey. *IEEE Internet Things J*. 2019;6(5):8076–94. doi: 10.1109/JIOT.2019.2920987.
13. Ferrag MA, Shu L, Yang X, Derhab A, Maglaras L. Security and privacy for green IoT-based agriculture: review, blockchain solutions, and challenges. *IEEE Access*. 2020;8:32031–53. doi: 10.1109/ACCESS.2020.2973178.
14. Yusuf SA, Khan A, Souissi R. Vehicle-to-everything (V2X) in the autonomous vehicles domain: a technical review of communication, sensor, and AI technologies for road user safety. *Transp Res Interdiscip Perspect*. 2024;23:100980. doi:10.1016/j.trip.2023.100980.
15. Cobb AD, Jalaian BA, Bastian ND, Russell S. Robust decision-making in the Internet of Battlefield Things using Bayesian neural networks. In: *Proceedings of the 2021 Winter Simulation Conference (WSC)*. IEEE; 2021. p. 1–12. doi: 10.1109/WSC52266.2021.9715532.
16. Sun W, Wang P, Xu N, Wang G, Zhang Y. Dynamic digital twin and distributed incentives for resource allocation in aerial-assisted Internet of Vehicles. *IEEE Internet Things J*. 2021;9(8):5839–52. doi: 10.1109/JIOT.2021.3058213.
17. Tanwar S, Tyagi S, Budhiraja I, Kumar N. Tactile internet for autonomous vehicles: latency and reliability analysis. *IEEE Wirel Commun*. 2019;26(4). doi: 10.1109/MWC.2019.1800553.
18. Vinita LJ, Vetrivel V. Federated learning-based misbehaviour detection on an emergency message dissemination scenario for the 6G-enabled Internet of Vehicles. *Ad Hoc Netw*. 2023;144:103153. doi: 10.1016/j.adhoc.2023.103153.
19. Liang J, Chen J, Zhu Y, Yu R. A novel intrusion detection system for vehicular ad hoc networks (VANETs) based on differences of traffic flow and position. *Appl Soft Comput*. 2019;75: 712–27. doi:10.1016/j.asoc.2018.12.001.
20. Gupta R, Tanwar S, Kumar N. B-IoMV: blockchain-based onion routing protocol for D2D communication in an IoMV environment beyond 5G. *Veh Commun*. 2022;33:10040. doi: 10.1016/j.vehcom.2021.100401.
21. Madadi B, Silvano AP, McPherson K, McCarthy J, Oorni R, Correia GHA. A connected and automated vehicle readiness framework to support road authorities for C-ITS services. *arXiv preprint arXiv:2311.01268*. 2023. doi:10.48550/arXiv.2311.01268.

22. He Y, Yu FR, Wei Z, Leung V. Trust management for secure cognitive radio vehicular ad hoc networks. *Ad Hoc Netw.* 2019;86:154–65. doi:10.1016/j.adhoc.2018.11.006.
23. Sharma R, Sharma TP, Sharma AK. TVDD: topology-based vehicular data dissemination scheme for stability optimisation in IoV. *Int J Sensor Netw.* 2021;35(4):243–57. doi:10.1504/IJSNET.2021.114751.
24. Liu B. A survey on trust modeling from a Bayesian perspective. *Wirel Pers Commun.* 2020;112(2):1205–27. doi:10.1007/s11277-020-07097-5.
25. Erdmann J. SUMO's lane-changing model. In: *Modeling mobility with open data: 2nd SUMO Conference 2014*, Berlin, Germany, May 15–16, 2014. Cham: Springer International Publishing; 2015. p. 105–123. doi:10.1007/978-3-319-15024-6_7.
26. Liu Y. VANET routing protocol simulation research based on NS-3 and SUMO. In: *2021 IEEE 4th International Conference on Electronics Technology (ICET)*. IEEE; 2021. p. 1073–6. doi: 10.1109/ICET51757.2021.9450921.
27. Sengupta P, Mehta A, Rana PS. Predictive maintenance of armoured vehicles using machine learning approaches. *arXiv preprint arXiv:2307.14453*. 2023. doi:10.48550/arXiv.2307.14453
28. Ali WA, Fanti MP, Roccotelli M, Ranieri L. A review of digital twin technology for electric and autonomous vehicles. *Appl Sci.* 2023;13(10):5871. doi:10.3390/app13105871.
29. Kumar R, Kumar P, Aljuhani A, Jolfaei A, Islam AKN, Mohammad N. Secure data dissemination scheme for digital twin empowered vehicular networks in Open RAN. *IEEE Trans Veh Technol.* 2023. doi: 10.1109/TVT.2023.3342127.
30. Gupta R, Jadav NK, Nair A, Tanwar S, Shahinzadeh H. Blockchain and AI-based secure onion routing framework for data dissemination in IoT environment underlying 6G networks. In: *2022 Sixth International Conference on Smart Cities, Internet of Things and Applications (SCIoT)*. IEEE; 2022. p. 1–6. doi: 10.1109/SCIoT56583.2022.9953671.

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In the current work, he has played a crucial role in developing the methodology by providing overall guidance and reviewing the simulation as well as experimental results.